

- [Image Classification with KerasHub](#)

```
from google.colab import files
uploaded = files.upload()
```

Choose Files 2 files

customer_data.csv(text/csv) - 47345 bytes, last modified: 5/2/2026 - 100% done
purchase_data.csv(text/csv) - 6977294 bytes, last modified: 5/2/2026 - 100% done
 Saving customer_data.csv to customer_data.csv
 Saving purchase_data.csv to purchase_data.csv

```
import os
print(os.listdir())
```

```
['.config', 'customer_data.csv', 'purchase_data.csv', 'sample_data']
```

```
import pandas as pd
```

```
purchase = pd.read_csv("purchase_data.csv")
customer = pd.read_csv("customer_data.csv")
```

```
print(purchase.head())
print(customer.head())
```

	order_id	customer_id	product_name	\
0	ODSHP00001	CS00001	Furniture	
1	ODSHP00002	CS00002	Dress	
2	ODSHP00003	CS00003	Furniture	
3	ODSHP00004	CS00004	Shoes	
4	ODSHP00005	CS00005	Rugs	

		description	price	discount	tax
0	Transform your space with this stylish and fun...		645.52	0.37	0.02
1	Look and feel your best with this elegant dress.		28.90	0.05	0.02
2	Transform your space with this stylish and fun...		3536.49	0.21	0.04
3	Step out in style with these fashionable shoes.		97.80	0.11	0.08
4	Add warmth and texture to your floors with the...		1785.38	0.13	0.00

	order_date	quantity	shipping_cost	shipping_date
0	26-05-2020	9	8.91	29-05-2020
1	29-12-2022	9	13.95	29-12-2022
2	06-03-2021	3	36.42	06-03-2021
3	11-03-2020	7	9.35	13-03-2020
4	12-03-2022	7	8.55	15-03-2022

	customer_id	first_name	last_name	gender	age	country	income
0	CS00001	Isadora	Porto	Female	19	Brazil	117196
1	CS00002	Hugo	Carreño	Male	33	Chile	49256
2	CS00003	René	Olivera	Male	65	Mexico	33434
3	CS00004	Luiz Henrique	Pinto	Male	55	Brazil	75302
4	CS00005	Leonardo	Monteiro	Male	19	Brazil	32280

```
import numpy as np
import matplotlib.pyplot as plt
```

```
from scipy import stats
```

```
purchase['Revenue'] = (purchase['price'] * purchase['quantity']) -
```

```
top_products = purchase.groupby('product_name')['Revenue'].sum().sort_values(ascending=False)
print(top_products)
```

```
product_name
Curtains          24685637.93
Wall Art          24405741.42
Lighting          23896470.84
Furniture         23697514.16
Kitchen Appliances 23637961.71
Name: Revenue, dtype: float64
```

Objective: The purpose of this analysis is to identify the products that contribute the most to total revenue.

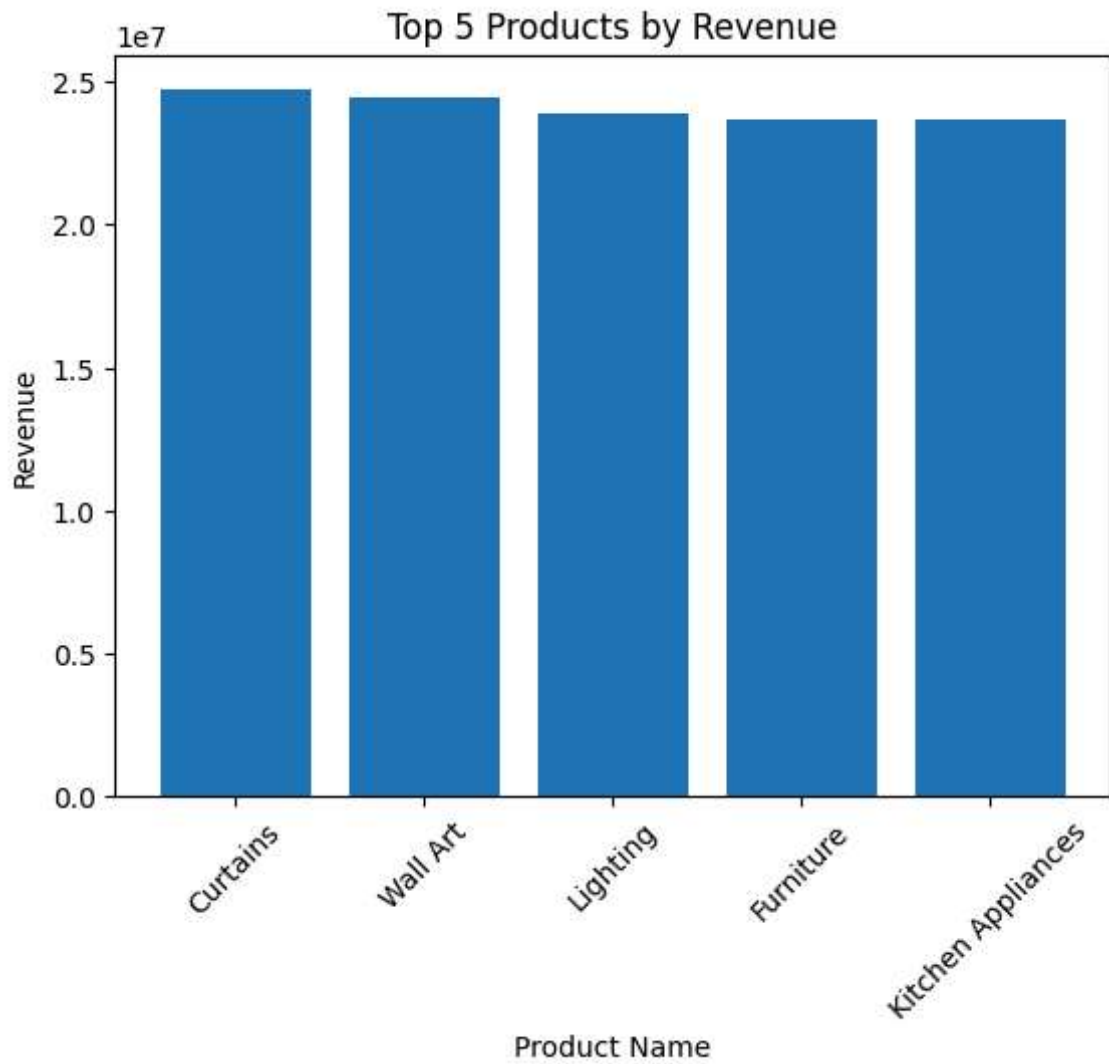
Explanation: Revenue was calculated using the formula: Revenue = (Price × Quantity) – Discount + Tax

After calculating revenue for each transaction, the total revenue for each product was aggregated. The top 5 products contributing the highest revenue were identified.

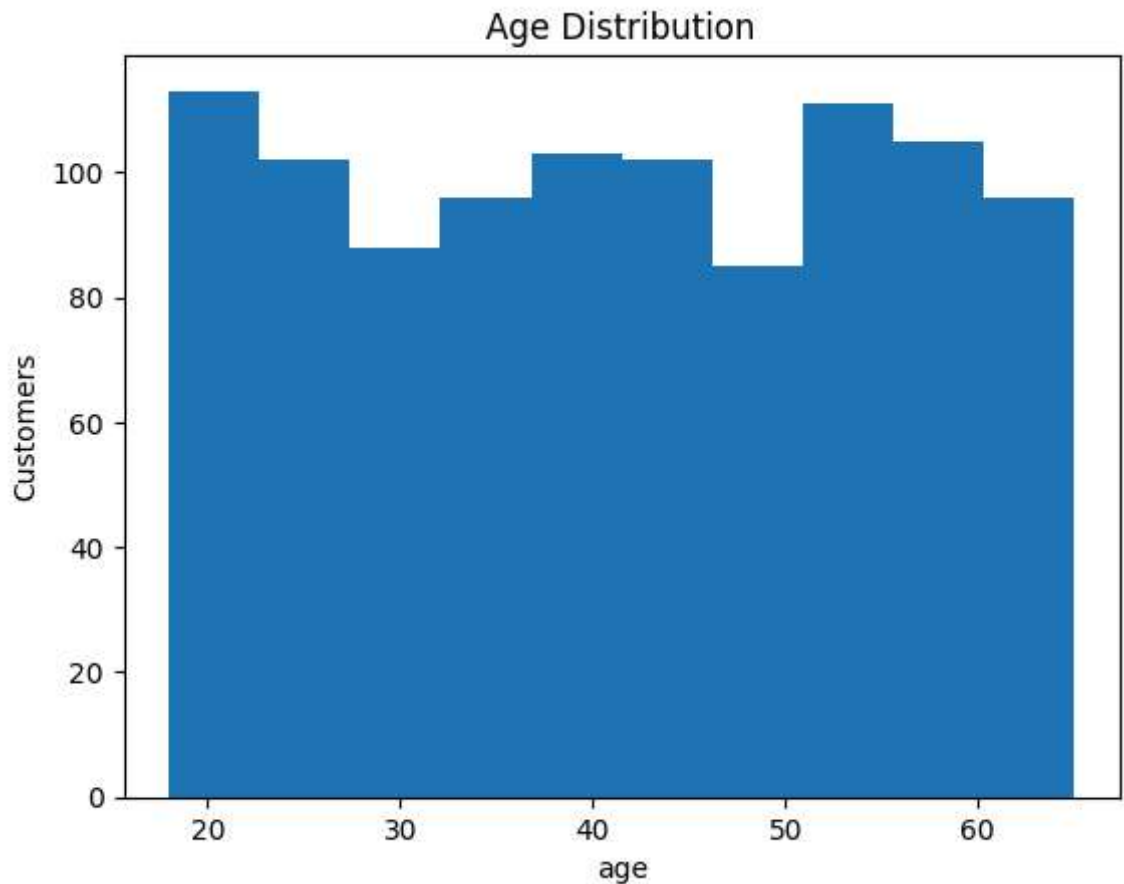
Interpretation: The bar chart shows that a small number of products generate a significant portion of total revenue. These products are the key drivers of business performance.

Conclusion: The company should focus on these high-performing products by optimizing pricing strategies, reducing unnecessary discounts, and ensuring their availability.

```
plt.figure()
plt.bar(top_products.index, top_products.values)
plt.title("Top 5 Products by Revenue")
plt.xlabel("Product Name")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.show()
```



```
plt.figure()
plt.hist(customer['age'], bins=10)
plt.title("Age Distribution")
plt.xlabel("age")
plt.ylabel("Customers")
plt.show()
```



Objective: To understand the age distribution of customers.

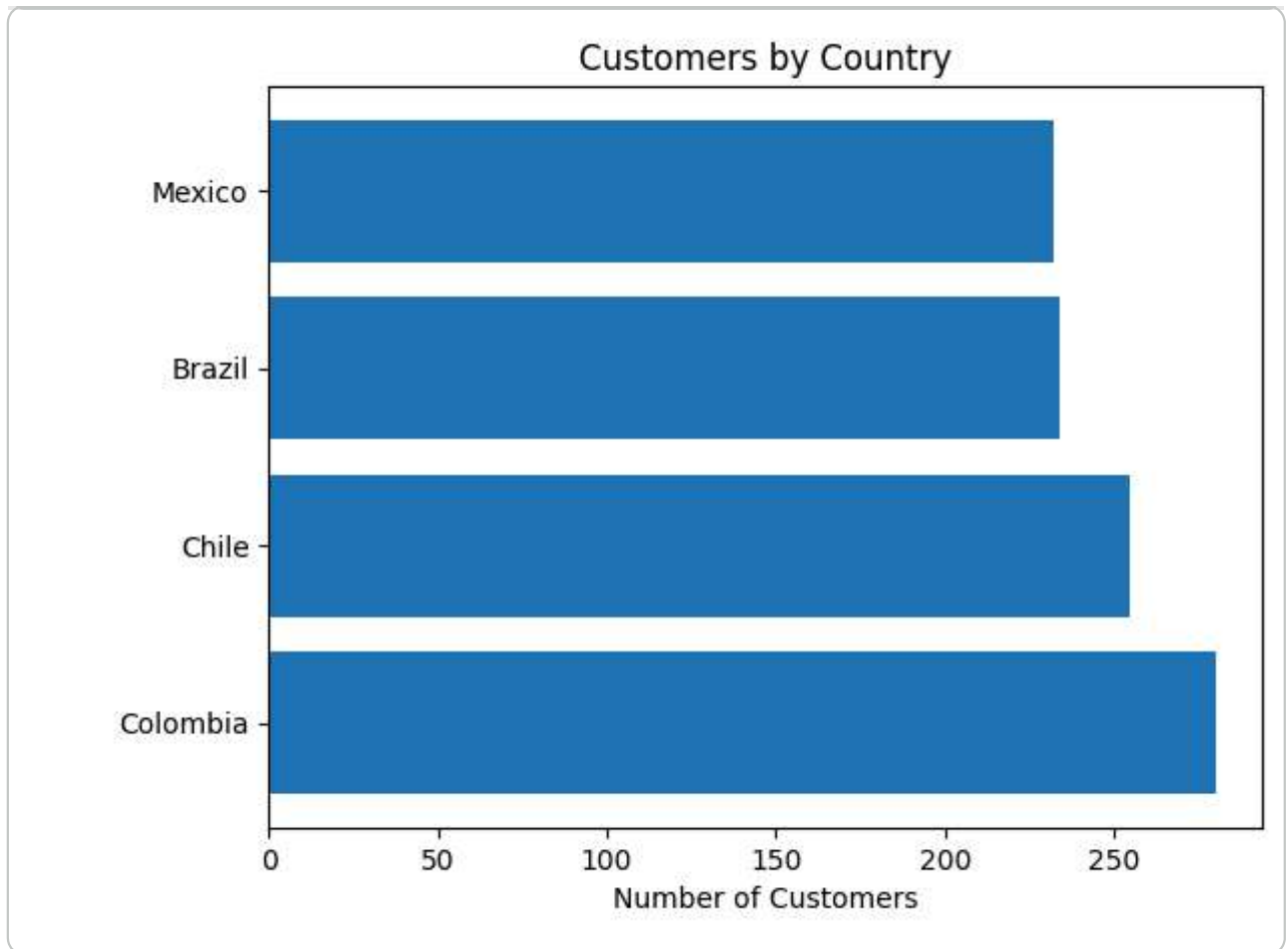
Explanation: A histogram was created to visualize how customers are distributed across different age groups.

Interpretation: The graph shows which age group has the highest number of customers. This helps in identifying the primary target audience.

Conclusion: The company can design marketing strategies tailored to the dominant age group to increase engagement and sales.

```
country_counts = customer['country'].value_counts()

plt.figure()
plt.barh(country_counts.index, country_counts.values)
plt.title("Customers by Country")
plt.xlabel("Number of Customers")
plt.show()
```



Objective: To identify which countries have the highest number of customers.

Explanation: A horizontal bar chart was used to display the number of customers from each country.

Interpretation: The countries with the highest number of customers represent the most important markets for the business.

Conclusion: The company should focus marketing efforts and business expansion strategies in these high-performing regions.

```
purchase['order_date'] = pd.to_datetime(purchase['order_date'])  
purchase['shipping_date'] = pd.to_datetime(purchase['shipping_date'])
```

%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True`

%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True`

```
purchase['Shipping_Days'] = (purchase['shipping_date'] - purchase['order_date']).dt.days
```

```
within_3 = purchase[purchase['Shipping_Days'] <= 3]['shipping_cost']  
after_3 = purchase[purchase['Shipping_Days'] > 3]['shipping_cost']
```

```
print("≤ 3 days:", within_3)
print("> 3 days:", after_3)
```

```
≤ 3 days: 27.561144983197313
> 3 days: 27.491996960162126
```

Objective: To analyze whether shipping time affects shipping cost.

Explanation: Orders were divided into two groups:

Orders shipped within 3 days Orders shipped after 3 days

The average shipping cost for both groups was calculated.

Interpretation: If the shipping cost is higher for delayed orders, it indicates inefficiencies in logistics.

Conclusion: The company should optimize its shipping process to ensure faster deliveries and reduce costs, improving both efficiency and customer satisfaction.

```
merged = pd.merge(purchase, customer, on='customer_id')
```

```
male_income = merged[merged['gender'] == 'Male']['income']
female_income = merged[merged['gender'] == 'Female']['income']

t_stat, p_value = stats.ttest_ind(male_income, female_income)

print("P-value:", p_value)
```

```
P-value: 0.03556694716265285
```

Objective: To determine whether there is a significant difference in income between male and female customers.

Hypotheses:

H0 (Null Hypothesis): There is no significant difference in average income between genders

H1 (Alternative Hypothesis): There is a significant difference

Explanation: An independent t-test was performed to compare the average income of male and female customers.

Interpretation:

If $p\text{-value} < 0.05 \rightarrow$ Reject H0 $\rightarrow$ Significant difference exists
If $p\text{-value} > 0.05 \rightarrow$ Accept H0 $\rightarrow$ No significant difference

Conclusion: Based on the result, the company can decide whether gender-based targeting strat

```
merged['Product_Category'] = merged['product_name'].apply(lambda x

import pandas as pd
contingency = pd.crosstab(merged['Product_Category'], merged['gender'])

chi2, p, dof, expected = stats.chi2_contingency(contingency)

print("P-value:", p)
P-value: 0.7948435792013748
```

Objective: To determine whether there is an association between product categories and customer gender